ML → AI Engineer Learning Roadmap v3 — Tailored to Your Current Skills (with links)

Starting Aug 11 | ~12 weeks total | 8-9 hrs/day, with built-in rest

Weeks 1-6: ML foundations (math, classical ML, basic deep learning, deployment basics)
Weeks 7-12: AI Engineer specialization (LLMs/RAG, backend APIs, Docker, cloud deployment) — added because "AI Engineer" roles today expect LLM application-building and software engineering skills on top of classical ML, not instead of it.

Why this version is different from v1

- v1 had you relearning Python/pandas/EDA you already know — cut.
- v1 crammed math + deep learning into single weeks with no slack — expanded.
- v1 had zero rest days across 30 days — added 1 buffer/rest day per week.
- v1 had no spaced review — added short recall sessions every few days on *previous* week's material, not just same-day revision.
- v2 stopped at classical ML + basic deep learning — v3 adds the LLM/RAG/backend/deployment layer needed to genuinely apply for "AI Engineer" titles, not just "Junior ML Engineer."

Week 1: Math Intuition + ML Foundations (Aug 11–17)

Skipped: Python/pandas/numpy/EDA basics — you already have this.

- **Math (3-4 days):**
  - 3Blue1Brown — Essence of Linear Algebra (playlist)
  - 3Blue1Brown — Essence of Calculus (playlist)
  - StatQuest — full channel & video index | StatQuest YouTube
- **Intro ML theory (3 days):**
  - Andrew Ng — Machine Learning Specialization (Coursera, audit free) — beginner-friendly, practical; use this as your primary source, not the old grad-level CS229 lectures
  - Kaggle Learn — Intro to Machine Learning (free micro-course) — a few hours, hands-on, great since you already know pandas/EDA
  - Primary channel pick: CampusX — 100 Days of Machine Learning — **CONFIRMED primary source** (Hindi explanations are working well for retention; stick with this throughout)
- **Day 7: Buffer/rest + light review**

Optional deeper pass later: Stanford CS229 full course, Autumn 2018 — grad-level and math-heavy, best as a second pass once the practical basics are solid, not a first introduction.

Mini project: Skip Titanic as a "learning" exercise — instead do a **quick model pass** (logistic regression + decision tree) on a dataset you're already comfortable EDA-ing, just to bridge into modeling. Don't over-invest time here.

Week 2: Classical ML — Core Focus (Aug 18–25, extended to 8 days)

This is your real starting line. No skipping. Extended by 1 day: merge/groupby/join were a genuine gap (not just refresher-level), so Day 8 is fully dedicated to closing that gap before starting the projects. Borrowed from Week 3's buffer day per the "if you fall behind" rule.

Topics: Pandas merge/join/concat/groupby (Day 8), Regression, Classification, Decision Trees, Ensembles (Random Forest, Boosting), Clustering, Dimensionality Reduction, Feature Engineering, Cross-Validation, Hyperparameter Tuning

Day-by-day plan: